1/30/2026

Create and Distribute a Torrent File in a Peer-to-Peer Environment

ASSIGNMENT - 2

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1. Introduction

Project Title: Creation and Distribution of a Torrent File in a Peer-to-Peer (P2P) Environment.

Abstract: This project illustrates decentralized file sharing using the BitTorrent protocol. A torrent metadata file is created for a chosen dataset, which is then distributed among peers. The system demonstrates seeding, downloading, and integrity verification without relying on a centralized file server.

Background: Traditional client–server systems depend on a single server for data delivery, often leading to scalability and availability issues. Peer-to-peer models address these limitations by enabling participants to share resources directly with one another.

Objectives: The project aims to achieve efficient file distribution, minimize server dependency, enhance scalability, and ensure reliable data integrity during transfer.

2. Distributed Systems Concepts

Explain the theory behind your project to show your academic understanding:

- **P2P Architecture:** Define unstructured vs. structured P2P networks.
- **Decentralization:** How the system functions without a central file server.
- **Scalability:** Why P2P systems handle high traffic better than traditional servers.
- **Data Integrity:** Mention how **SHA-1/SHA-256 hashing** is used to verify file pieces.

3. Project Objectives

The objectives of this project are listed below:

- To study the working of Peer-to-Peer architectures
- To create torrent metadata files for data sharing
- To implement decentralized file transfer mechanisms
- To improve download performance through parallel sharing
- To understand practical distributed system concepts

4.Background and Motivation

Modern distributed applications require reliable and efficient mechanisms to share large datasets. Centralized systems often face challenges such as limited scalability, heavy server load, and single points of failure. These drawbacks motivate the use of decentralized approaches.

This project focuses on implementing a torrent-based file sharing mechanism in a peer-to-peer environment. By enabling peers to exchange file segments directly, the system improves performance, fault tolerance, and overall resource utilization.

5. Challenges & Solutions

- **Peer Churn:** What happens when a seeder goes offline?
- **Network Issues:** Handling NAT traversal or firewall blocks.
- **Synchronization:** Ensuring pieces are requested in a way that doesn't cause bottlenecks.

6. Project Workflow

The overall workflow of the system is as follows:

1. Torrent Creation: A .torrent file is generated from the source data, containing metadata and tracker information.
2. Seeding: The original peer makes the complete file available to the network.
3. Peer Discovery: New peers contact the tracker to locate available seeders and leechers.
4. Distribution: File pieces are exchanged among peers until the full file is reconstructed.

7. Operational Workflow

The process begins when a user creates a torrent metadata file for the selected content. This metadata is registered with a tracker, allowing other peers to discover sources for the data.

Peers request file fragments based on availability and priority. After obtaining all fragments, a peer reconstructs the original file and may continue to share it, thereby strengthening the network.

8.System Architecture

The system architecture follows a decentralized Peer-to-Peer model with minimal reliance on centralized components. The primary elements of the system include torrent creator, tracker, seeders, and leechers.

The tracker acts as a coordination service, while actual file data transfer occurs directly between peers. Each peer functions both as a client and a server, allowing efficient sharing of resources. This architecture enhances scalability and ensures high availability even when some nodes fail.

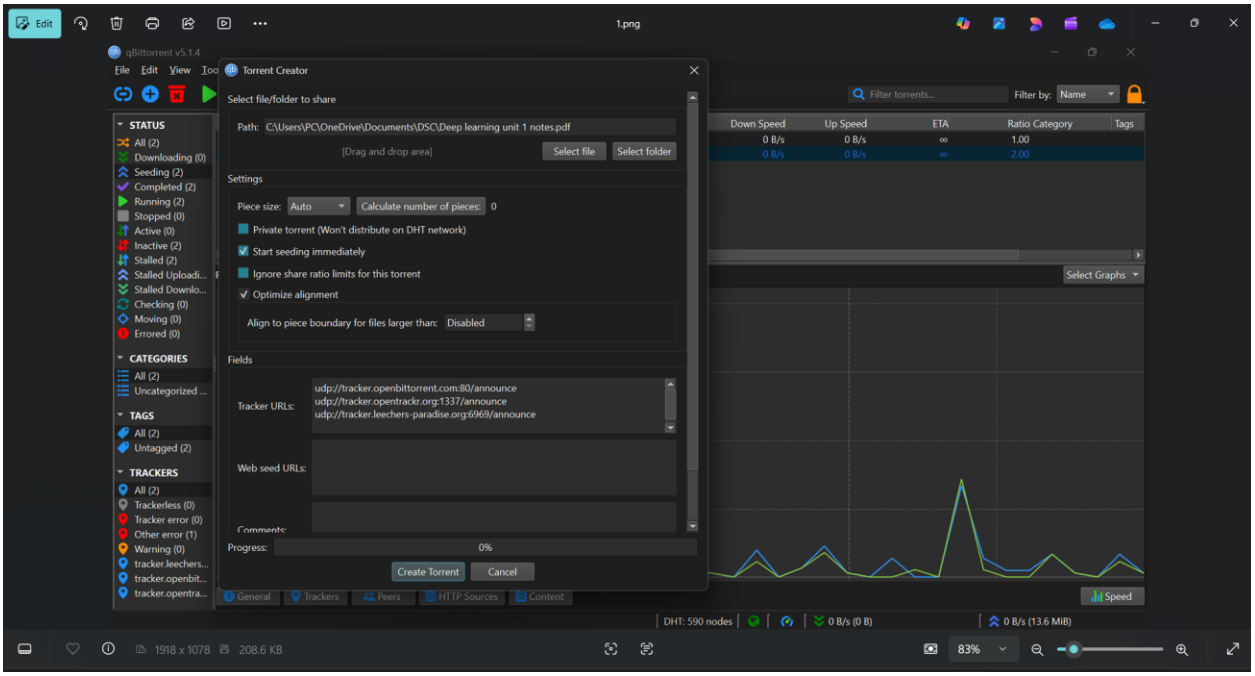
9.Testing and Results

- **Test Cases:** Show what happened when 2 peers joined, then 5, then 10.
- **Performance Metrics:** Download speeds vs. number of seeders.
- **Integrity Check:** Screenshots or logs showing that the file was verified successfully after download.

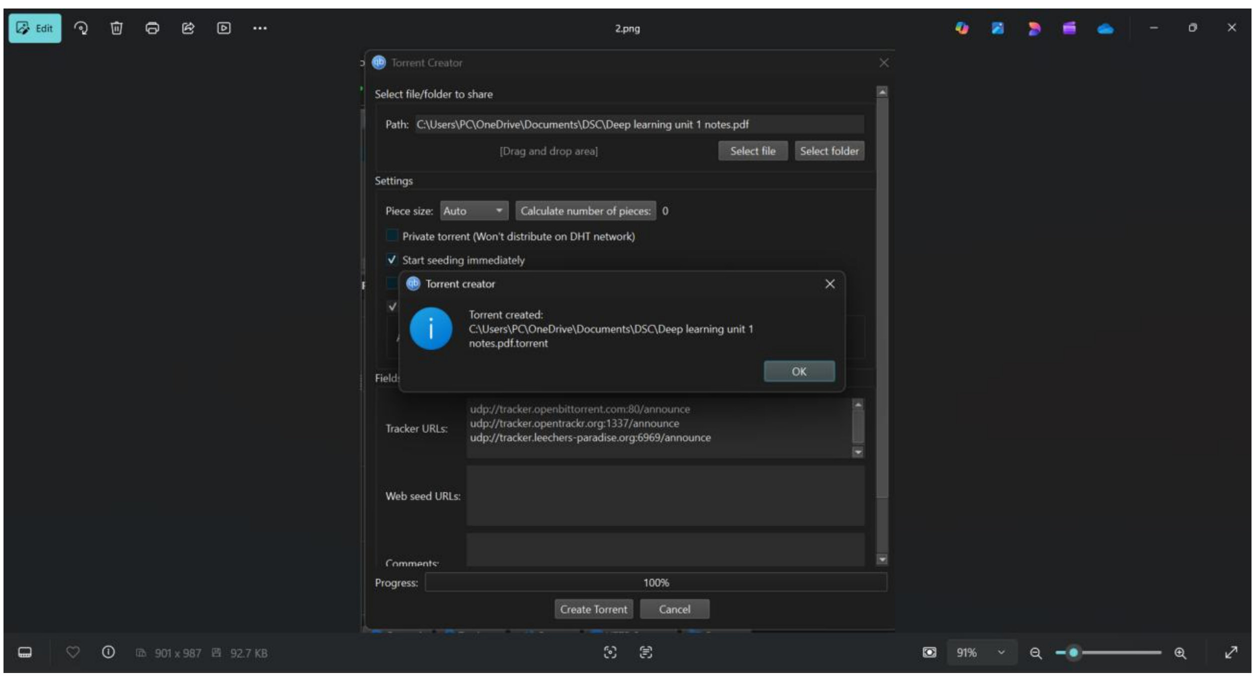
10.Outputs

The implemented system effectively demonstrates torrent-based file sharing in a distributed network. Observations indicate reduced reliance on centralized servers, improved transfer speeds, and efficient utilization of network resources

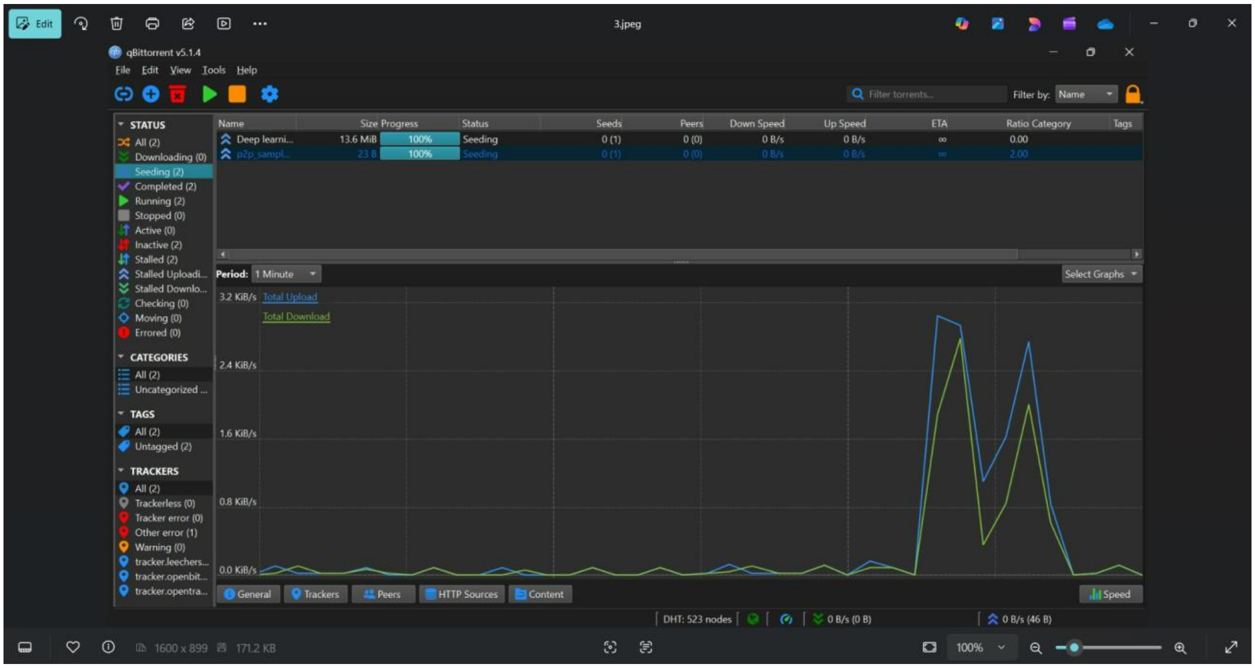
SENDER:



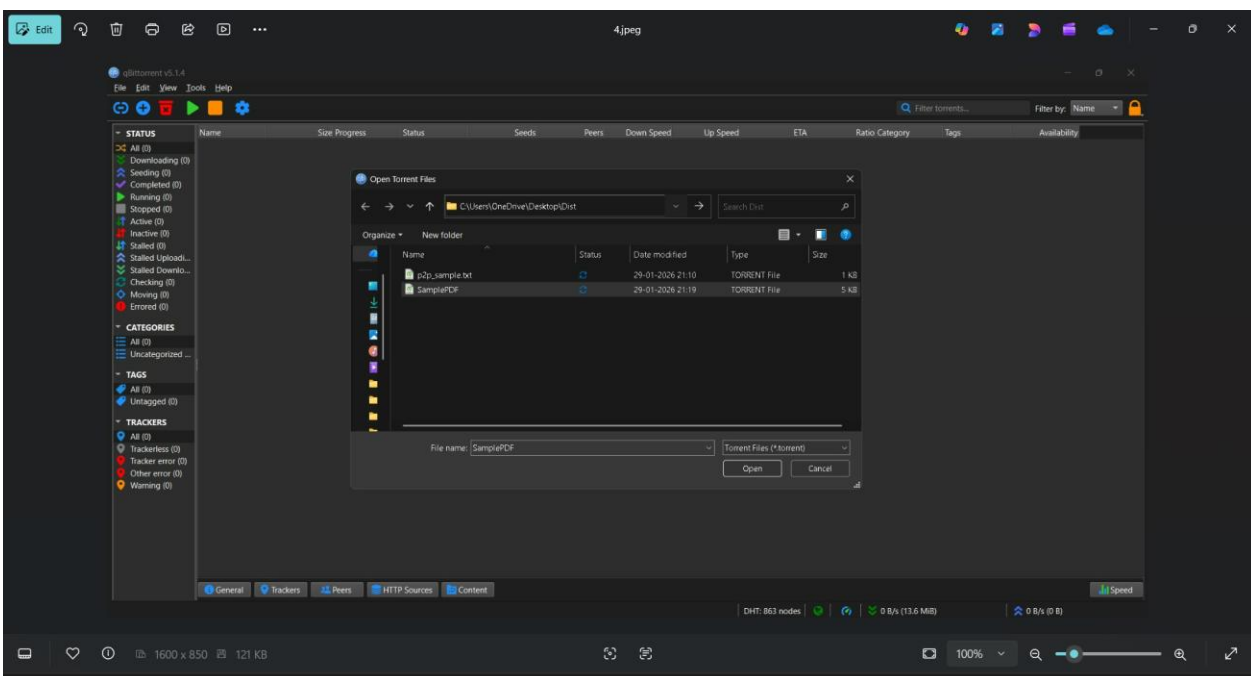
1.png
Torrent Creator
Select file/folder to share
Path: C:\Users\PC\OneDrive\Documents\DSC\Deep learning unit 1 notes.pdf
[Drag and drop area]
Select file
Select folder
Settings
Piece size: Auto
Calculate number of pieces: 0
Private torrent (Won't distribute on DHT network)
Start seeding immediately
Ignore share ratio limits for this torrent
Optimize alignment
Align to piece boundary for files larger than: Disabled
Fields
Tracker URLs:
udp://tracker.openbittorrent.com:80/announce
udp://tracker.opentrackr.org:1337/announce
udp://tracker.leechers-paradise.org:6969/announce
Web seed URLs:
Comments:
Progress: 0%
Create Torrent
Cancel

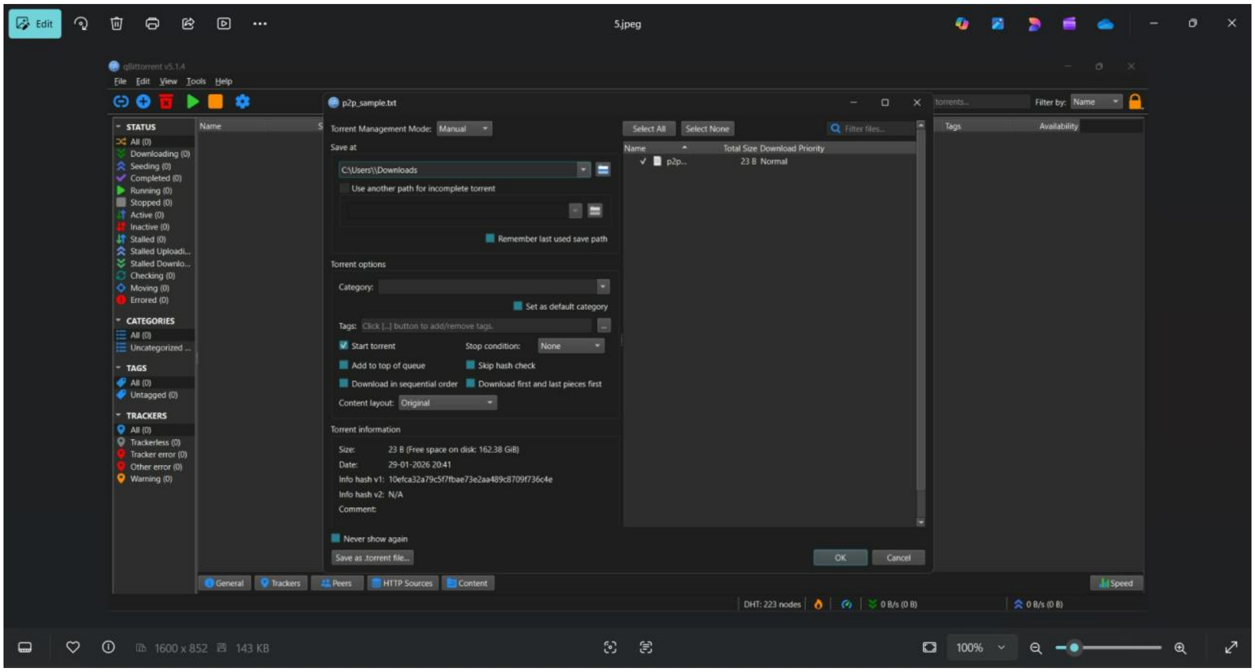


2.png
Torrent Creator
Select file/folder to share
Path: C:\Users\PC\OneDrive\Documents\DSC\Deep learning unit 1 notes.pdf
[Drag and drop area]
Select file
Select folder
Settings
Piece size: Auto
Calculate number of pieces: 0
Private torrent (Won't distribute on DHT network)
Start seeding immediately
Torrent creator
Torrent created:
C:\Users\PC\OneDrive\Documents\DSC\Deep learning unit 1 notes.pdf.torrent
OK
Tracker URLs:
udp://tracker.openbittorrent.com:80/announce
udp://tracker.opentrackr.org:1337/announce
udp://tracker.leechers-paradise.org:6969/announce
Web seed URLs:
Comments:
Progress: 100%
Create Torrent
Cancel

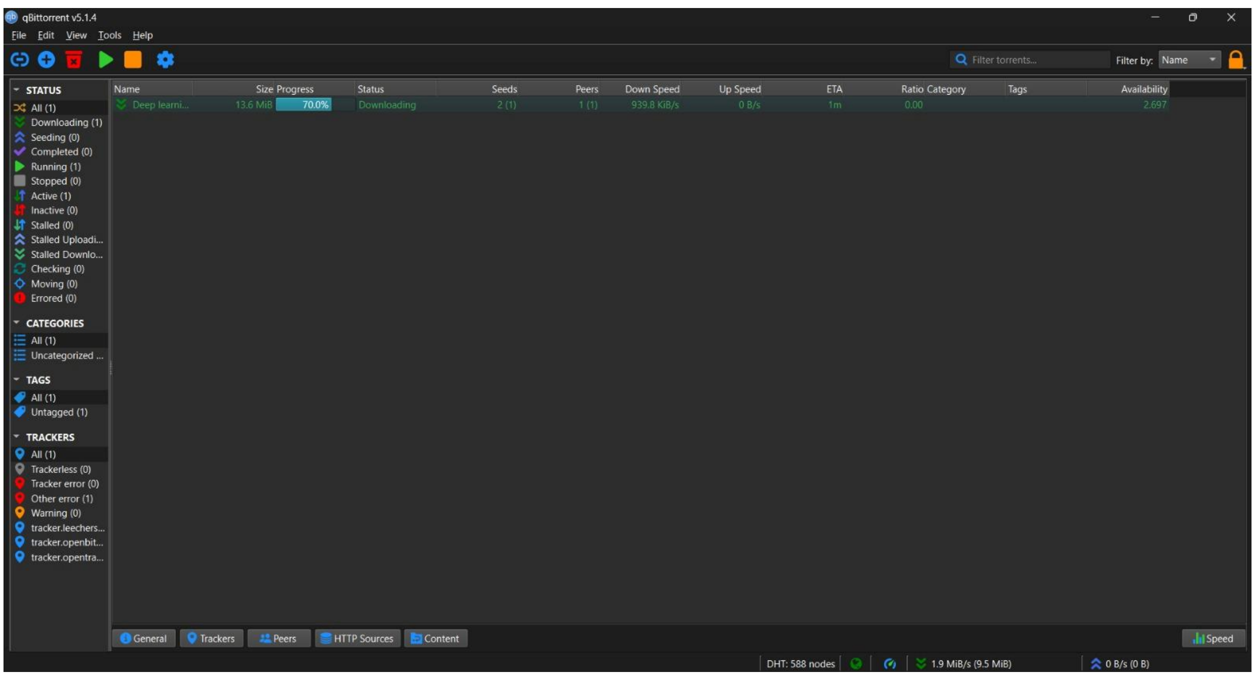


RECEIVER:





Edit
5.jpeg
qBittorrent v5.1.4
File Edit View Tools Help
p2p_sample.txt
Torrent Management Mode: Manual
Save at
C:\Users\\Downloads
Use another path for incomplete torrent
Remember last used save path
Torrent options
Category:
Set as default category
Tags: Click [...] button to add/remove tags.
Start torrent
Stop condition: None
Add to top of queue
Skip hash check
Download in sequential order
Download first and last pieces first
Content layout: Original
Torrent information
Size: 23 B (Free space on disk: 162.38 GiB)
Date: 29-01-2026 20:41
Info hash v1: 10efca32a79c5f7fbae73e2aa489c8709f736c4e
Info hash v2: N/A
Comment:
Never show again
Save as .torrent file...
Select All
Select None
Filter files...
Name
Total Size Download Priority
p2p...
23 B Normal
OK
Cancel
STATUS
All (0)
Downloading (0)
Seeding (0)
Completed (0)
Running (0)
Stopped (0)
Active (0)
Inactive (0)
Stalled (0)
Stalled Uploadi...
Stalled Downlo...
Checking (0)
Moving (0)
Errored (0)
CATEGORIES
All (0)
Uncategorized ...
TAGS
All (0)
Untagged (0)
TRACKERS
All (0)
Trackerless (0)
Tracker error (0)
Other error (0)
Warning (0)
General
Trackers
Peers
HTTP Sources
Content
Speed
DHT: 223 nodes
0 B/s (0 B)
0 B/s (0 B)
1600 x 852
143 KB
100%



qBittorrent v5.1.4
File Edit View Tools Help
Filter torrents...
Filter by: Name
STATUS
All (1)
Downloading (1)
Seeding (0)
Completed (0)
Running (1)
Stopped (0)
Active (1)
Inactive (0)
Stalled (0)
Stalled Uploadi...
Stalled Downlo...
Checking (0)
Moving (0)
Errored (0)
CATEGORIES
All (1)
Uncategorized ...
TAGS
All (1)
Untagged (1)
TRACKERS
All (1)
Trackerless (0)
Tracker error (0)
Other error (1)
Warning (0)
tracker.leechers...
tracker.openbit...
tracker.opentra...
Name Size Progress Status Seeds Peers Down Speed Up Speed ETA Ratio Category Tags Availability
Deep learni... 13.6 MiB 70.0% Downloading 2 (1) 1 (1) 939.8 KiB/s 0 B/s 1m 0.00 2.697
General
Trackers
Peers
HTTP Sources
Content
Speed
DHT: 588 nodes
1.9 MiB/s (9.5 MiB)
0 B/s (0 B)

11. Conclusion

In conclusion, torrent-based P2P systems provide a robust and scalable solution for large-scale file sharing. The project effectively applies distributed system principles such as decentralization, concurrency, and reliability, making it a practical implementation for modern data distribution needs.